

LISBOA GEOFISICO, Portugal

WMO: 085350

Lat: 38.7191N

Lon: 9.1497W

Elev: 78

StdP: 100.39

Time Zone: 0.00 (EUW)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.7	6.8	-4.6	2.6	9.3	-2.0	3.2	10.1	10.3	14.0	9.3	12.8	3.8	40	0.422

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.9	33.2	20.1	31.3	19.7	29.6	19.3	22.3	29.3	21.2	27.8	20.6	26.7	3.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.5	15.3	23.4	19.4	14.3	22.5	18.7	13.6	22.1	65.5	29.3	61.8	27.9	59.4	26.8	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.9	7.9	7.1	3.8	37.3	1.7	2.0	2.5	38.7	1.5	39.8	0.6	40.9	-0.7	42.4		
			0.9	23.5	2.1	1.9	-0.6	24.9	-1.8	26.0	-2.9	27.1	-4.4	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	12.0	12.6	14.5	16.2	18.8	21.5	22.9	23.8	22.0	19.3	14.9	12.4
	DBStd	4.86	2.16	2.04	2.40	2.65	3.10	2.95	2.58	2.54	2.30	2.55	2.38	2.25
	HDD10.0	18	7	4	1	0	0	0	0	0	0	0	1	6
	HDD18.3	895	196	161	122	77	30	3	0	0	0	17	106	184
	CDD10.0	2795	70	77	139	186	274	346	400	427	360	289	147	80
	CDD18.3	631	0	0	2	12	46	99	142	169	111	48	2	0
	CDH23.3	3902	0	0	12	57	306	641	962	1199	557	167	1	0
	CDH26.7	1281	0	0	1	4	86	223	341	441	155	30	0	0
Wind	WSAvg	3.5	3.2	3.5	3.6	3.7	3.7	3.6	4.0	3.5	3.1	3.1	3.1	3.3
Precipitation	PrecAvg	735	105	92	72	62	47	18	3	5	30	94	121	101
	PrecMax	1189	353	298	267	163	156	98	43	58	129	262	373	383
	PrecMin	416	0	0	1	5	0	0	0	0	0	0	2	4
	PrecStd	206	76	67	59	40	40	22	7	11	32	74	86	91
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.5	20.0	24.9	27.3	31.8	34.5	36.2	36.1	33.2	29.8	22.6	18.7
		MCWB	13.5	13.5	15.3	17.0	18.4	21.1	21.4	20.4	19.1	18.1	16.3	15.4
	2%	DB	17.0	18.2	22.4	24.9	29.1	31.6	32.8	33.0	30.3	26.8	20.7	17.4
		MCWB	13.6	12.9	14.0	16.6	18.1	20.1	20.3	20.3	18.6	17.8	16.2	15.0
	5%	DB	16.0	17.0	20.2	23.1	26.8	29.3	30.2	31.0	28.3	25.0	19.5	16.6
		MCWB	13.4	12.7	13.6	15.9	17.5	19.4	19.8	19.9	18.7	17.6	15.7	14.5
	10%	DB	15.3	15.9	18.4	20.9	24.5	27.0	28.0	28.9	26.4	23.4	18.4	15.8
		MCWB	13.0	12.3	13.2	14.8	16.7	18.8	19.2	19.6	18.3	17.2	15.2	13.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.9	15.7	17.2	19.2	21.4	24.0	23.6	24.1	22.0	21.0	19.1	16.8
		MCDB	16.5	17.0	21.3	24.0	26.9	29.7	29.9	29.3	28.9	24.1	20.2	17.5
	2%	WB	15.1	14.9	15.7	17.6	19.6	21.4	21.9	22.1	20.8	20.0	18.0	15.9
		MCDB	16.0	16.0	19.1	23.1	25.5	28.2	29.9	29.1	26.4	22.7	19.1	16.8
	5%	WB	14.4	14.2	14.9	16.3	18.3	20.3	20.8	21.1	20.1	19.2	17.0	15.1
		MCDB	15.4	15.7	17.4	21.4	24.4	27.2	27.7	27.6	25.0	22.2	18.7	16.2
	10%	WB	13.6	13.4	14.2	15.3	17.3	19.4	20.0	20.4	19.4	18.5	16.0	14.2
		MCDB	14.8	15.2	17.0	19.7	22.8	25.6	26.4	26.6	23.7	21.4	17.7	15.5
Mean Daily Temperature Range	5% DB	MDBR	5.8	6.7	7.3	7.9	8.5	9.0	9.7	9.9	8.6	7.1	5.9	5.4
		MCDBR	6.3	7.9	10.0	10.6	11.7	12.5	12.8	12.3	11.2	9.4	6.7	5.7
		MCWBR	3.9	4.4	5.0	5.6	5.7	6.1	5.6	5.7	5.0	4.4	3.9	3.9
	5% WB	MCDBR	5.0	6.3	7.7	9.4	10.0	11.0	11.0	10.6	9.0	7.0	5.5	5.1
		MCWBR	4.1	4.5	4.6	5.4	5.7	6.0	5.5	5.5	4.6	3.9	4.0	4.1
Clear-Sky Solar Irradiance	taub	0.328	0.341	0.370	0.387	0.395	0.399	0.384	0.397	0.396	0.362	0.338	0.328	
	taud	2.516	2.476	2.404	2.350	2.358	2.365	2.400	2.373	2.388	2.503	2.527	2.515	
	Ebn at Noon	849	884	888	891	886	879	890	870	849	848	830	820	
	Edh at Noon	78	92	108	121	123	122	117	117	109	88	76	73	
All-Sky Solar Radiation	RadAvg	2.25	3.25	4.42	5.68	6.82	7.36	7.49	6.80	5.32	3.66	2.50	2.03	
	RadStd	0.28	0.38	0.46	0.46	0.44	0.38	0.29	0.26	0.34	0.30	0.31	0.23	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page